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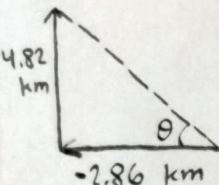
Excellent!

PHY201 Final Exam on chapter(s): 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15

Name: _____

These questions assess your understanding of the material from Chapters 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 of OpenStax College Physics. Please write your name on the sheets of paper that you use and show all your work.

1. Out of curiosity, you want to conduct some geographic measurements of your house and your friend Nick's. To get to Nick's, you have to walk 2.86 km west and then 4.82 km north. What is the compass direction of a line connecting your house to his? Give the angle and the specific direction. Draw it if necessary.



$$\theta = \tan^{-1}\left(\frac{\Delta y}{\Delta x}\right)$$

$$\theta = \tan^{-1}\left(\frac{4.82 \text{ km}}{2.86 \text{ km}}\right) = 59.32^\circ$$

ANSWER: 59.32° North of West ✓

2. If a marathon runner averages 11 mi/h, how long does it take her to run a 18.5-mi marathon?

$$\vec{v}_{\text{avg}} = \frac{\Delta \vec{x}}{t} \rightarrow t = \frac{\Delta \vec{x}}{\vec{v}_{\text{avg}}}$$

$$t = \frac{18.5 \text{ mi}}{11 \frac{\text{mi}}{\text{h}}} = 1.68 \text{ h}$$

ANSWER: 1.68 h ✓

3. To test the idea of surviving a free-falling elevator, Stephen conducts an experiment using a string-attached plank and a programmable toy hare. His makeshift elevator falls 17.0 m and the hare jumps 0.0005 s before impact. Previous tests show that these toys can endure impacts below 6.29 m/s. What is the minimum initial speed needed for the hare's vertical jump to make it impact the ground at a safe speed? (Note: The plank's mass is significantly greater than the toy's mass.)

ANSWER: _____

+2

4. Laika the moon dog jumps into the air to reach a height of 3.95 m. How long is Laika in the air? Laika is on the Moon and so her free-fall acceleration is 1.625 m/s^2 . Note: Laika was a stray part-terrier mix from Moscow who, in 1957, became the first living creature to enter space and orbit Earth, clearing the celestial trail for human space exploration.

$$t_1 = t_u + t_d = 2t_u$$

$$v_f^2 = v_i^2 + 2a\Delta y \rightarrow v_i = \sqrt{-2a\Delta y}$$

$$v_f = v_i + at_u \rightarrow t_u = \frac{-v_i}{a} = \frac{-\sqrt{-2a\Delta y}}{a}$$

$$t_u = \frac{-\sqrt{-2(-1.625 \frac{\text{m}}{\text{s}^2})(3.95 \text{ m})}}{-1.625 \frac{\text{m}}{\text{s}^2}} = 2.205 \text{ s}$$

$$t_1 = 2(2.205 \text{ s}) = \boxed{4.41 \text{ s}}$$

ANSWER: 4.41 s

5. Ian throws a laptop horizontally from a bridge with an initial speed of 33.3 m/s . Find the laptop's speed when it enters the water 9.7 seconds later. You may ignore air resistance.

$$v_{x_i} = v_{x_f}$$

$$v_{y_f} = v_{y_i} + a_y t$$

$$v_f = \sqrt{v_{x_f}^2 + v_{y_f}^2} = \sqrt{v_{x_i}^2 + (a_y t)^2}$$

$$v_f = \sqrt{(33.3 \frac{\text{m}}{\text{s}})^2 + (-9.8 \frac{\text{m}}{\text{s}^2}(9.7 \text{ s}))^2} = \boxed{100.72 \frac{\text{m}}{\text{s}}}$$

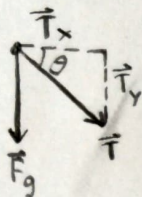
ANSWER: 100.72 $\frac{\text{m}}{\text{s}}$

6. Alexis aims her arrow towards a target located 200.8 m away. The target's bull's eye aligns with the height from which Alexis releases the arrow. Launched at an angle with an initial speed of 47.9 m/s , the arrow passes a thin tree branch positioned directly between Alexis and the target. The branch is 108.4 m away from Alexis and is positioned 9.5 meters above the arrow's launch point. What is the vertical distance at which the arrow passes by the branch? A positive value indicates it passed above the branch, while a negative value indicates it passed below.

ANSWER: _____

7. Rachel goes outside on a summer morning with her tea and sees a perfectly horizontal strand of spider web. She watches for a while and eventually a spider climbs to the middle of the strand, making the web bend at an angle. Rachel gets a protractor and measures the angle to be 18° below the horizontal. At this point, invested in her experiment, Rachel takes the spider and puts it on a scale, measuring its mass to be 530 mg, then returns it to its strand of web. What is the tension in the web when the spider is sitting on it?

ANSWER: _____



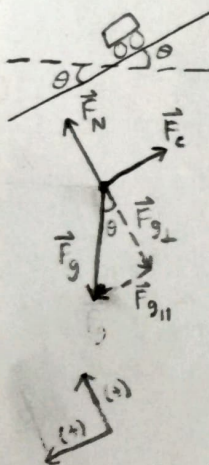
8. Charles is trying to keep a 125.8-kg wagon at rest on a frictionless inclined plane, but despite his best efforts, he can only apply a force of 439 N to try to stop the motion. If the plane makes an angle of 62.8° with the horizontal, what is the acceleration of the wagon down the plane?

ANSWER: $5.23 \frac{m}{s^2}$

$$\Sigma F_{||} = -F_c + F_{g||} = ma_{||} = F_g \sin \theta - F_c$$

$$a_{||} = \frac{mg \sin \theta - F_c}{m}$$

$$a_{||} = \frac{(125.8 \text{ kg})(9.8 \frac{m}{s^2}) \sin(62.8^\circ) - 439 \text{ N}}{125.8 \text{ kg}} = \boxed{5.23 \frac{m}{s^2}}$$



9. Dylan nudges a 4.8-kg block of wood so that it just begins moving down a long, wooden ramp angled at 17° above the horizontal. The block of wood must travel 74.5 m to reach the bottom. How long does it take for the block of wood to reach the bottom? Do not neglect friction. Note that for wood on wood, $\mu_s = 0.5$ and $\mu_k = 0.3$.

ANSWER: 52.67 s

$$\Sigma F_{||} = F_{g||} - f_k = ma_{||} = F_g \sin \theta - \mu_k F_N$$

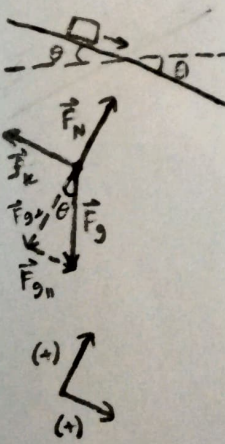
$$\Sigma F_{\perp} = F_N - F_{g\perp} = m a_{\perp} \rightarrow F_N = F_{g\perp} = F_g \cos \theta = mg \cos \theta$$

$$\vec{s}_{||f} = \vec{s}_{||i} + \vec{v}_{||i} t + \frac{1}{2} \vec{a}_{||} t^2 \rightarrow \vec{a}_{||} = \frac{\vec{s}_{||f}}{\frac{1}{2} t^2} \rightarrow a_{||} = \frac{2 s_{||f}}{t^2}$$

$$m \cdot \frac{2 s_{||f}}{t^2} = mg \sin \theta - \mu_k mg \cos \theta = g (\sin \theta - \mu_k \cos \theta)$$

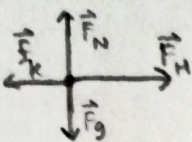
$$t = \sqrt{\frac{2 s_{||f}}{g (\sin \theta - \mu_k \cos \theta)}}$$

$$t = \sqrt{\frac{2(74.5 \text{ m})}{(9.8 \frac{m}{s^2})(\sin(17^\circ) - 0.3 \cos(17^\circ))}} = \boxed{52.67 \text{ s}}$$



+2

10. A team of thirteen huskies pulls Ian and his sled across snow at 1.7 m/s. Ian and his loaded sled have a combined mass of 330 kg and the average mass of each dog is 20.8 kg. If each dog can pull with a force of 150 N, what is the magnitude of the acceleration of the sled? Note that for waxed wood on snow, $\mu_s = 0.14$ and $\mu_k = 0.1$.



$$\Sigma F_x = F_H - f_k = m a_x = F_H - \mu_k F_N$$

$$\Sigma F_y = F_N - F_g = m a_y \rightarrow F_N = F_g = mg$$

$$a_x = \frac{F_H - \mu_k mg}{m}$$

$$a_x = \frac{13(150 \text{ N}) - 0.1(330 \text{ kg})(9.8 \frac{\text{m}}{\text{s}^2})}{330 \text{ kg}} = \boxed{4.93 \frac{\text{m}}{\text{s}^2}}$$

ANSWER: $4.93 \frac{\text{m}}{\text{s}^2}$

11. A 17-kg child is riding a playground merry-go-round that is rotating at 6.2 revolutions per minute. What centripetal force must she exert to stay on if she is 7.3 m from the center of the merry-go-round?

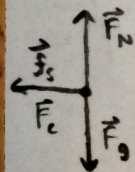
$$\frac{6.2 \text{ rev}}{1 \text{ min}} \left| \frac{2\pi \text{ rad}}{1 \text{ rev}} \right| \left| \frac{1 \text{ min}}{60 \text{ s}} \right| = 0.649 \frac{\text{rad}}{\text{s}} = \omega$$

ANSWER: 52.31 N

$$F_c = m a_c = m r \omega^2$$

$$F_c = (17 \text{ kg})(7.3 \text{ m})(0.649 \frac{\text{rad}}{\text{s}})^2 = \boxed{52.31 \text{ N}}$$

12. Chloe drives her 1030-kg car around a 290-m radius unbanked curve at 31.8 m/s (about 71 mph). Chloe's mass is 61 kg. Find the minimum static coefficient of friction between the tires and the road that prevents the car from slipping.



$$f_s = F_c = m a_c = m \cdot \frac{v^2}{r} \leq \mu_s F_N$$

$$\Sigma F_y = F_N - F_g = m a_y \rightarrow F_N = F_g = mg$$

$$\frac{m v^2}{r} \leq \mu_s m g \rightarrow \mu_s \geq \frac{v^2}{r g}$$

$$\mu_s \geq \frac{(31.8 \frac{\text{m}}{\text{s}})^2}{(290 \text{ m})(9.8 \frac{\text{m}}{\text{s}^2})} = \boxed{0.36}$$

ANSWER: 0.36

f3

13. Katherine places a 0.11-kg toy car next to a spring on a track. She moves the car backward to compress the spring 29.7 cm, then releases the car, propelling it upward along a curved track. How fast is the car going after it rises to a height of 2.127 m above the starting position? Since Katherine greased the axles of the toy car, assume there is no friction. Katherine's previous experiments indicate that the spring has a force constant of 100 N/m.

$$KE_i + PE_i + W_{nc} = KE_f + PE_f \rightarrow \frac{1}{2} k \Delta x^2 = \frac{1}{2} m v_f^2 + mgh_f$$

ANSWER:

$$8.93 \frac{m}{s}$$

$$v_f = \sqrt{\frac{\frac{1}{2} k \Delta x^2 - mgh_f}{\frac{1}{2} m}}$$

$$v_f = \sqrt{\frac{\frac{1}{2} (100 \frac{N}{m}) (0.297 m)^2 - (0.11 kg) (9.8 \frac{m}{s^2}) (0.02127 m)}{\frac{1}{2} (0.11 kg)}} = 8.93 \frac{m}{s}$$

14. Matthew is designing a skate park and wants to build a curved ramp (quarterpipe) that leads skaters a distance of 7.3 m to the top of a straight ramp. Matthew plans that skaters will drop into the quarterpipe from rest and launch off the straight ramp with a speed of 1.0 m/s. If the height of the quarterpipe is 2.6 m, what must the height of the straight ramp be? Experimental studies in skate science show that the average friction force per mass on skaters as they skate across this particular surface is 0.57 N/kg.

$$KE_i + PE_i + W_{nc} = KE_f + PE_f$$

ANSWER: 2.12 m

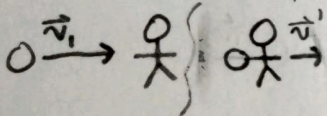
$$W_{nc} = W_f = -f d_s$$

$$mgh_i - f d_s = \frac{1}{2} m v_f^2 + mgh_f \rightarrow h_f = \frac{mgh_i - f d_s - \frac{1}{2} m v_f^2}{mg}$$

$$h_f = \frac{(1 kg) (9.8 \frac{m}{s^2}) (2.6 m) - (0.57 N) (7.3 m) - \frac{1}{2} (1 kg) (1 \frac{m}{s})^2}{(1 kg) (9.8 \frac{m}{s^2})} = 2.12 m$$

15. Brian, a 95-kg ice-hockey goalie who is originally at rest, catches a 0.17-kg ice-hockey puck that is traveling at a velocity of 36.9 m/s. Calculate Brian's speed after he catches the puck.

ANSWER: 6.59 $\frac{cm}{s}$



$$\Delta \vec{p} = 0 = \vec{p}' - \vec{p} \rightarrow \vec{p} = \vec{p}'$$

$$m_i \vec{v}_i = m' \vec{v}' \rightarrow v' = \frac{m_i v_i}{m'}$$

$$v' = \frac{(0.17 kg) (36.9 \frac{m}{s})}{0.17 kg + 95 kg} = 0.0659 \frac{m}{s}$$

- ✓ 16. Christopher, a professional heavyweight boxer, delivers a forceful punch with his fist and arm, combining for an effective punching mass of 3.1 kg. His fist strikes a heavy bag at a velocity of 9.3 m/s and comes to a halt 9.4 ms after initial contact. Determine the magnitude of the average net force that Christopher exerts on the heavy bag.

$$\Delta \vec{p} = \vec{p}' - \vec{p} = -m_i \vec{v}_i = \vec{F}_{avg} \Delta t \rightarrow \vec{F}_{avg} = \frac{-m_i \vec{v}_i}{t}$$

ANSWER: 3067.02 N

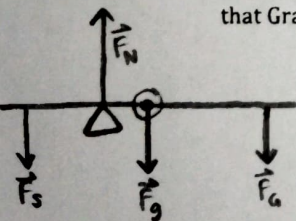
$$F_{avg} = \frac{(3.1 \text{ kg})(9.3 \frac{\text{m}}{\text{s}})}{0.0094 \text{ s}} = \boxed{3067.02 \text{ N}}$$

17. Abigail is designing a set to be shown in a play. A 79-kg actor stands on a 53-kg platform that is suspended by two ropes, one on its left side, and another on its right side. The left rope is attached to a weight of mass m_L through a frictionless pulley; the right rope is attached to a weight of mass m_R through another frictionless pulley. When the weights are released, the platform must descend with an acceleration of 0.1 m/s^2 without rotating. Abigail needs to make the platform 2.7-m long and the actor must stand 0.5-m from the left side. What must m_R be?

ANSWER: _____

- ✓ 18. Sean is building a seesaw for his daughter, Grace, and plans on making it 12 ft long. He plans to sit on one side, 5.6 ft away from the center of the seesaw, and his daughter will sit on the other side, 5.6 ft away from the center of the seesaw. Sean has weighed himself, his daughter, and the seesaw and learned that his weight is 175 lbs, his daughter's weight is 46 lbs, and the weight of the seesaw is 10.2 lbs. He wants to place the fulcrum so that the seesaw is perfectly balanced, so that Grace can cause a rotation in the seesaw just as easily as himself. Where must the fulcrum be placed?

ANSWER: 3.12 ft from center on Sean's side



$$\sum \tau = -\tau_a - \tau_N + \tau_s = I\alpha$$

$$r_N F_N = r_s F_s - r_a F_a$$

$$\sum F_y = F_N - F_s - F_g - F_a = m a_y \rightarrow F_N = F_s + F_g + F_a$$

$$r_N = \frac{r_s F_s - r_a F_a}{F_s + F_g + F_a}$$

$$r_N = \frac{(5.6 \text{ ft})(175 \text{ lbs}) - (5.6 \text{ ft})(46 \text{ lbs})}{175 \text{ lbs} + 10.2 \text{ lbs} + 46 \text{ lbs}} = \boxed{3.12 \text{ ft}}$$

F3

19. Dylan is spinning a large grindstone by placing his hand on its edge and exerting a force through part of a revolution. He exerts a force of 312 N through a rotation of 1.13 rad (65°). He exerts the force perpendicular to the grindstone's 44-cm radius and the effects of friction are negligible. What is the grindstone's final angular velocity if its mass is 55 kg? Consider the grindstone to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

$$\Sigma W = \Delta KE = KE_f - KE_i = \frac{1}{2} I \omega^2 = \Sigma \tau \theta = \tau \theta$$

ANSWER: 7.63 rad/s

$$\frac{1}{2} \cdot \frac{1}{2} M R^2 \omega^2 = r F \theta \rightarrow \omega = \sqrt{\frac{r F \theta}{\frac{1}{4} M R^2}}$$

$$\omega = \sqrt{\frac{(0.44 \text{ m})(312 \text{ N})(1.13 \text{ rad})}{\frac{1}{4}(55 \text{ kg})(0.44 \text{ m})^2}} = \boxed{7.63 \frac{\text{rad}}{\text{s}}}$$

20. Ethan, a deep-sea fisherman, hooks a big fish that swims away from the boat, pulling the fishing line from the fishing reel. The whole system is initially at rest and the fishing line unwinds from the reel at a radius of 4.1 cm from the reel's axis of rotation. The reel is given an angular acceleration of 140 rad/s² for 1.0 s. How many meters of fishing line come off the reel in this time?

$$\theta_f = \theta_i + \omega_i t + \frac{1}{2} \alpha t^2 \rightarrow \Delta \theta = \frac{1}{2} \alpha t^2$$

ANSWER: 2.87 m

$$\Delta x = r \Delta \theta = r \cdot \frac{1}{2} \alpha t^2$$

$$\Delta x = \frac{1}{2} (0.041 \text{ m})(140 \frac{\text{rad}}{\text{s}^2})(1 \text{ s})^2 = \boxed{2.87 \text{ m}}$$

21. Calculate the maximum buoyant force that water could exert on a ship with a mass of 2.5×10^8 kg if the ship displaces 7.0×10^6 m³ of water.

ANSWER: 6.86×10^{10} N

$$F_b = m_w g$$

$$\rho_w = \frac{m_w}{V_s} \rightarrow m_w = V_s \rho_w$$

$$F_b = V_s \rho_w g$$

$$F_b = (7 \times 10^6 \text{ m}^3)(1000 \frac{\text{kg}}{\text{m}^3})(9.8 \frac{\text{m}}{\text{s}^2}) = \boxed{6.86 \times 10^{10} \text{ N}}$$

22. Intravenous (IV) infusions are usually made with the help of the gravitational force. Assuming that the density of the fluid being administered is 1.00 g/mL, at what height should the IV bag be placed above the entry point so that the fluid only just enters the vein if the blood pressure in the vein is 13.5 mmHg above atmospheric pressure? Recall that 1.0 mmHg = 133.32 Pa.

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$$

$$P_{atm} + \rho gh_1 = P_{atm} + P_A \rightarrow h_1 = \frac{P_A}{\rho g}$$

$$h_1 = \frac{1799.82 \frac{\text{kg}}{\text{m}^3} \cdot \text{m} \cdot \text{s}^{-2}}{(1000 \frac{\text{kg}}{\text{m}^3})(9.8 \frac{\text{m}}{\text{s}^2})} = \boxed{0.18 \text{ m}}$$

ANSWER: 0.18 m

$$\frac{13.5 \text{ mmHg} | 133.32 \text{ Pa}}{1 \text{ mmHg}} = 1799.82 \text{ Pa}$$

23. John turns a hose faucet on and after 8.9 seconds, sees water emerge from his 8-m-long garden hose of diameter 2.1 cm. John then turns it off momentarily to attach a nozzle of diameter 8.6 mm to the end, then turns the water back on. He then uses the hose to fill up a 3-L bucket. What is the speed of the water that leaves the nozzle?

ANSWER: _____

24. The aorta pumps blood from the heart to the rest of the body. The flow rate of blood through Brandon's aorta is 14.4 L/min and Brandon's aorta has a radius of 13 mm. Recall that blood flows through smaller blood vessels known as capillaries, which have an average diameter of 8.0 μm . The speed of blood in the capillaries is about 0.28 mm/s. Calculate the number of capillaries in Brandon's blood circulatory system.

$$\frac{14.4 \text{ L} | 1 \text{ m}^3}{1 \text{ min} | 1000 \text{ L} | 60 \text{ s}} = 2.4 \times 10^{-4} \frac{\text{m}^3}{\text{s}}$$

ANSWER: 1.71×10^{10} capillaries

$$n_1 A_1 \bar{v}_1 = n_2 A_2 \bar{v}_2 = n_1 Q = n_2 \pi r_2^2 \bar{v}_2 \rightarrow n_2 = \frac{n_1 Q}{\pi r_2^2 \bar{v}_2}$$

$$n_2 = \frac{(1)(2.4 \times 10^{-4} \frac{\text{m}^3}{\text{s}})}{\pi (4 \times 10^{-6} \text{ m})^2 (0.00028 \frac{\text{m}}{\text{s}})} = \boxed{1.71 \times 10^{10} \text{ capillaries}}$$

25. Daniel inflates his bicycle tire to an absolute pressure of 0.71 MPa (a gauge pressure of about 88 psi) at a temperature of 18.5 °C. What is the tire's gauge pressure after the temperature has risen to 33.4°C? Assume there are no leaks and the volume of the tire does not change.

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} = \frac{P_g + P_{atm}}{T_2} \rightarrow P_g = \frac{P_1 T_2}{T_1} - P_{atm}$$

ANSWER: 644947.93 Pa

$$P_g = \frac{(0.71 \times 10^6 \text{ Pa})(33.4 + 273.15) \text{ K}}{(18.5 + 273.15) \text{ K}} - 101325 \text{ Pa} = \boxed{644947.93 \text{ Pa}}$$

26. Julia is a design engineer for a lightbulb company. She manufactures a gas-filled incandescent light bulb so that the gas inside the bulb is at atmospheric pressure when the bulb has a temperature of 7°C. What is the pressure in the light bulb when it rises to 298°C? The coefficient of volume expansion for the glass light bulb is $28 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$.

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} = \frac{P_2 (V_1 + \Delta V)}{T_2} = \frac{P_2 (V_1 + \beta V_1 \Delta T)}{T_2}$$

ANSWER: 204904.68 Pa

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_1 (1 + \beta \Delta T)}{T_2} \rightarrow P_2 = \frac{P_1 T_2}{T_1 (1 + \beta \Delta T)}$$

$$P_2 = \frac{(101325 \text{ Pa})(298 + 273.15) \text{ K}}{(7 + 273.15) \text{ K} (1 + (28 \times 10^{-6} \text{ } ^\circ\text{C}^{-1})(298 - 7) ^\circ\text{C})} = \boxed{204904.68 \text{ Pa}}$$

27. Grace puts a 0.39-kg aluminum pot on the stove and fills it with 0.15 liters of water. Both the pot and the water are initially at 15.1°C. How much heat is required to bring them both to 80.9°C? The specific heat value of water is 4186 J/kg °C and for aluminum it is 900 J/kg °C.

$$Q_T = Q_A + Q_W = m_A c_A \Delta T + m_W c_W \Delta T$$

ANSWER: 64411.62 J

$$Q_T = \Delta T (m_A c_A + V_W \rho_W c_W)$$

$$Q_T = (80.9 - 15.1) ^\circ\text{C} \left((0.39 \text{ kg})(900 \frac{\text{J}}{\text{kg} \cdot ^\circ\text{C}}) + (0.00015 \text{ m}^3)(1000 \frac{\text{kg}}{\text{m}^3})(4186 \frac{\text{J}}{\text{kg} \cdot ^\circ\text{C}}) \right) = \boxed{64411.62 \text{ J}}$$

+3

28. David is producing heat at a rate of about 235 W while doing light exercise. At what rate must water evaporate from his body to get rid of the produced energy? The latent heat of vaporization of water is 2430 kJ/kg at body temperature, 37°C.

$$Q = mL_v \rightarrow \frac{Q}{t} = \frac{mL_v}{t} = P \rightarrow \frac{m}{t} = \frac{P}{L_v}$$

ANSWER: $9.67 \times 10^{-5} \frac{\text{kg}}{\text{s}}$

$$\frac{m}{t} = \frac{235 \frac{\text{J}}{\text{s}}}{2430000 \frac{\text{J}}{\text{kg}}} = \boxed{9.67 \times 10^{-5} \frac{\text{kg}}{\text{s}}}$$

29. Spontaneous heat transfer from hot to cold is an irreversible process. Calculate the total change in entropy if 4430 J of heat transfer occurs from a hot reservoir at 544 K to a cold reservoir at 225 K, assuming that there is no temperature change in either reservoir.

$$\Delta S_T = \Delta S_H + \Delta S_C = \frac{Q_H}{T_H} + \frac{Q_C}{T_C}$$

ANSWER: $11.55 \frac{\text{J}}{\text{K}}$

$$\Delta S_T = \frac{-4430 \text{ J}}{544 \text{ K}} + \frac{4430 \text{ J}}{225 \text{ K}} = \boxed{11.55 \frac{\text{J}}{\text{K}}}$$

30. Find the increase in entropy as 6.8 kg of ice, originally at 1.5°C, melts to form water at 1.5°C at a particular pressure. The latent heat of fusion for water is 334 kJ/kg.

$$\Delta S = \frac{Q}{T} = \frac{mL_f}{T}$$

ANSWER: $8.27 \frac{\text{kJ}}{\text{K}}$

$$\Delta S = \frac{(6.8 \text{ kg})(334 \frac{\text{kJ}}{\text{kg}})}{(1.5 + 273.15) \text{ K}} = \boxed{8.27 \frac{\text{kJ}}{\text{K}}}$$

f3